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CSS6334
DIGITAL & COMPUTER FORENSICS

Lecture 1: Introduction to Forensics

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Learning Objectives

After studying this chapter, you should be able to:

- The definition and importance of digital forensics
- Understand different types of digital evidence
- Understand the skills required to become a digital forensics investigator
- Understand the history of digital forensics
- Understand agencies in the Malaysia, and internationally involved in digital forensics investigations

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The Definition and Importance of Computer Forensics

- Digital forensics is the retrieval, analysis, and use of digital evidence in a civil or criminal investigation.
- Any medium that can store digital files is a potential source of evidence for a computer forensics investigator.
- Computer forensics is a science because of the accepted practices used for acquiring and examining the evidence and its admissibility in court.
- *Forensically sound* means that during the acquisition of digital evidence and throughout the investigative process the evidence must remain in its original state.
- Moreover, everyone who has been in contact with the evidence must be accounted for and documented in the Chain of Custody form.

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What Is Computer Forensics?

The use of analytical and investigative techniques to identify, collect, examine, and preserve evidence/information that is magnetically stored or encoded

Objective is to recover, analyze, and present computer-based material in such a way that it can be used as evidence in a court of law

Applies to all the domains of a typical IT infrastructure: User Domain, Workstation Domain, LAN Domain, LAN-to-WAN Domain, WAN Domain, Remote Access Domain, System/Application Domain

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What Is Computer Forensics? (Cont.)

The seven domains of a typical IT infrastructure.

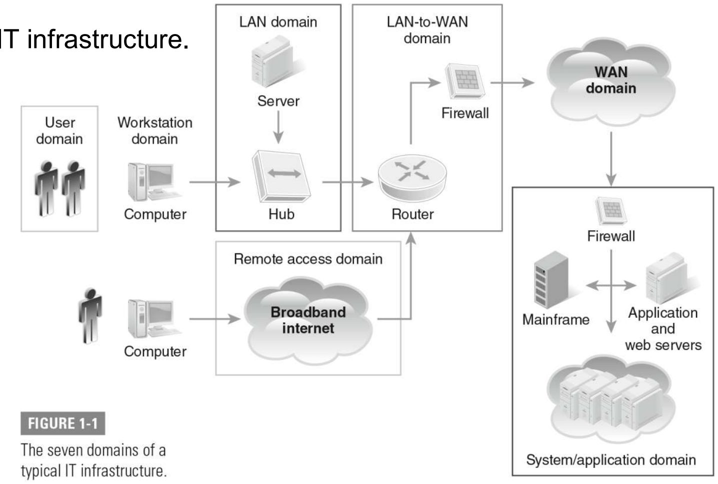

FIGURE 1-1
The seven domains of a typical IT infrastructure.

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Using Scientific Knowledge

- Important to understand and apply scientific methods and processes and have knowledge of the relevant scientific disciplines, including scientific knowledge of the field
- Requires a thorough understanding of:
  - Computer hardware
  - Operating system running on that device
    - May include smartphones and routers
  - At least the basics of computer networks
- Stay up-to-date
  - Keep learning what is there, where it is stored, and how that information may be used by computer user and computer criminal alike

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Using Scientific Knowledge (Cont.)

Collecting	• How you collect evidence determines if that evidence is admissible in a court
Analyzing	• Putting together the data you have and finding out what sort of picture is revealed
Presenting	• The expert report • Expert testimony

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Understanding the Field of Digital Forensics

- Field is changing very rapidly: Emerging standards provide clear, codified methods for conducting a forensic examination
- Entities involved in and actively using computer forensics:
  - The military
  - Government agencies
  - Law firms
  - Criminal prosecutors
  - Academia
  - Data recovery firms
  - Corporations
  - Insurance companies
  - Individuals

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What Is Digital Evidence?

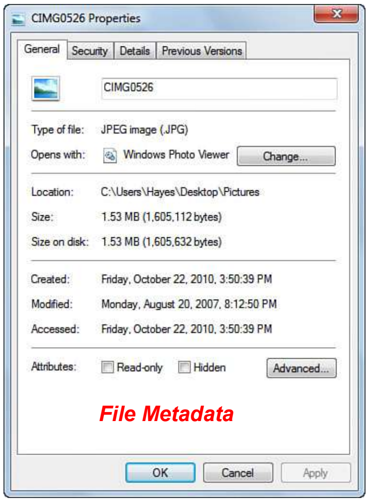
- Information that has been processed and assembled so that it is relevant to an investigation and supports a specific finding or determination
- Raw information is not, in and of itself, evidence
- Data must be relevant to a case in order to be evidence

- Real
- Documentary
- Testimonial
- Demonstrative

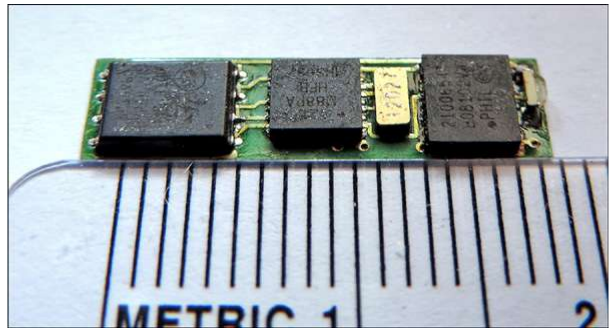
Different Types of Forensic Evidence

- Practically every type of file can be recovered using computer forensics.
  - Physical storage
  - Email is arguably the most important type of digital evidence
  - Network/ Internet
  - Images/ Audio/ Video
  - Software/ Live system
  - Websites visited and Internet searches
  - Cellphones
  - IoT Devices

Different Types of Forensic Evidence (Cont.)



File Metadata



Skimming Devices Used to Capture Data Stored on the Magnetic Stripe of an ATM Card

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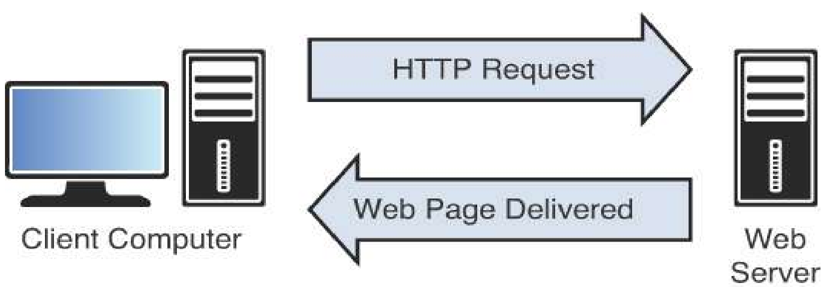
Different Types of Forensic Evidence (Cont.)



Communication Between a Client and a Web Server

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General Guidelines

- Maintain chain of custody
- Do not touch suspect drive
- Create a document trail
- Secure evidence

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What Skills Must a Computer Forensics Investigator Possess?

- Computer Science Knowledge
- Legal Expertise
- Communication Skills
- Linguistic Abilities
- Continuous Learning
- Appreciation for Confidentiality

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Role of Forensic Practitioners

- Collect, preserve, and analyze physical and digital evidence.
- Maintain chain of custody to ensure evidence integrity.
- Prepare reports and testify in court as expert witnesses.
- Collaborate with law enforcement, legal professionals, and scientists.
- Contribute to advancements in forensic methods and technologies.

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Obscured Information and Anti-Forensics

Obscured information

- May be secured by encryption, hidden using steganographic software, compressed, or in a proprietary format
- Cybercriminals sometimes obscure information to deter forensic examination
- Companies use certain manipulation and storage techniques to protect business-sensitive information

Anti-forensics

- Attackers may use techniques to intentionally conceal their identities, locations, and behavior
- Examples:
  - Data destruction
  - Data hiding
  - Data transformation
  - File system alteration

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The Importance of Computer Forensics (In General)

- The Bureau of Labor Statistics has recognized the importance of computer forensics and security.
- It estimates that between 2008 and 2028 job opportunities will increase by 12 percent.
- The increase in employment opportunities will result from an increase in criminal activity on the Internet, such as identity theft, spamming, email harassment, and illegal downloading of copyrighted materials.
- Skilled computer forensics examiners also have job opportunities within private investigation firms.
- Jobs for computer forensics investigators are available in many sectors of the economy, propelled by the digitization of our personal information.

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The Importance of Computer Forensics in Malaysia

1. **Cybercrime Growth -** Rising cases of cyber fraud, data breaches, and online scams. Computer forensics supports investigation and prosecution of digital crimes.
2. **Legal & Regulatory Compliance -** Supports enforcement under acts such as: Computer Crimes Act 1997, Personal Data Protection Act (PDPA), Evidence Act 1950 (digital evidence admissibility)
3. **National & Organizational Security -** Helps protect government, business, and financial systems from cyber threats. Ensures data integrity and system resilience.
4. **Digital Evidence & Justice -** Recovers and preserves digital evidence for court cases. Assists law enforcement, lawyers, and auditors in criminal investigations.
5. **Workforce Demand & Career Relevance -** Growing need for certified forensic analysts and cybersecurity professionals. Key skill area in Malaysia's digital economy initiatives (e.g., MyDIGITAL).

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The History of Computer Forensics

- 1981: IBM introduced the 5150 PC.
- 1984: The FBI established the Magnetic Media Program, later known as CART.
- 1984: The National Center for Missing and Exploited Children (NCMEC) was founded.
- 1985 HTCIA was founded in CA.
- 1986: The USSS established the Electronic Crimes Task Force (ECTF).
- 1986: Congress passed the Computer Fraud and Abuse Act.
- 1993: The first International Conference on Computer Evidence took place.
- 1994: Congress passed the Crime Bill, and the USSS began working on crimes against children.

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The History of Computer Forensics (Cont.)

- 1994: Mosaic Netscape, the first graphical web browser, was released.
- 1995: The International Organization on Computer Evidence (IOCE) was formed.
- 1996: USSS founded the New York Electronic Crimes Task Force (ECTF).
- 1999: The First Regional Computer Forensics Laboratory (RCFL) was established in San Diego.
- 2000: The IRS Criminal Investigation Division (IRS-CID) began using ILook.
- 2001: The USA PATRIOT Act and USSS were directed to establish ECTFs nationwide.
- 2001: INTERPOL developed a database of exploited children (ICAID).

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The History of Computer Forensics (Cont.)

- 2002: The Department of Homeland Security (DHS) was formed.
- 2003: The PROTECT Act was passed to fight against child exploitation.
- 2003: Fusion centers were established.
- 2007: The National Computer Forensics Institute (NCFI) was established.
- 2008: The formation of an INTERPOL Computer Forensics Analysis Unit was approved.
- 2009: The first European ECTF was formed (Italy).
- 2010: The second European ECTF was formed (United Kingdom).

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The History of Computer Forensics (Cont.)

- 2011: The PATRIOT Sunsets Extension Act was enacted.
- 2013: Edward Snowden fled to Russian following a classified data leak at the NSA.
- 2015: The trial of The Silk Road ends with the conviction of its creator, Ross Ulbricht.
- 2015: The Cybersecurity Information Sharing Act was enacted.
- 2016: Introduction of the Investigatory Powers Act in the U.K.
- 2017: APFS introduced, by Apple, with iOS 10.3.
- 2018: Introduction of the General Data Protection Regulation.

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The History of Computer Forensics (Cont.)

1980s: The Advent of the Personal computer

- The first electronic bulletin boards emerged and facilitated communication between hackers.
- Subsequently, hacking groups, such as the Legion of Doom in the United States, emerged.
- The 1983 film *War Games* introduced the public to the concept of hacking with a personal computer to gain access to government computers.
- In 1984, Eric Corley (with the handle Emmanuel Goldstein) published *2600: The Hacker Quarterly* , which facilitated the exchange of hacking ideas. In 1984, the FBI established the Magnetic Media Program, which subsequently became known as the Computer Analysis and Response Team (CART), and was responsible for computer forensics examinations.
- In 1984, the U.S. Congress established the National Center for Missing and Exploited Children (NCMEC).

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The History of Computer Forensics (Cont.)

1990s: The Impact of the Internet

- Web browsers prompted a massive migration of computers to the Internet.
- In 1998, the Defense Reform Initiative Directive #27 directed the U.S. Air Force to establish the joint Department of Defense Computer Forensics Laboratory, which would be responsible for counterintelligence, and criminal and fraud computer evidence investigations.
- The IRS Criminal Investigation Division Electronic Crimes Program funded Elliott Spencer to develop a computer forensics tool known as ILook.
- In 1994, Congress mandated that the USSS apply its forensic and technical knowledge to criminal investigations connected to missing and exploited children.
- In 2001, the USA PATRIOT Act mandated that the United States Secret Service expand its successful New York Electronic Crimes Task Force and establish ECTFs nationwide.

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The History of Computer Forensics in Malaysia

Early Legal & Institutional Foundations

- The Computer Crimes Act 1997 (CCA 1997) was enacted to address misuse of computers in Malaysia.
- The Digital Signature Act 1997 provided a framework for electronic signatures, helping shape the legal basis for digital-evidence handling.
- As digital devices and networks grew, the need for forensic capabilities to investigate cybercrime and preserve digital evidence became clearer.

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The History of Computer Forensics in Malaysia (Cont.)

Establishment of Key Forensic Laboratory

- In 1998, the CyberSecurity Malaysia (CSM), through its predecessor, established a Digital Forensics Laboratory (DFL) under what was then the National ICT Security & Emergency Response Centre (NISER).
- The laboratory officially announced digital forensics services around 2002
- As digital devices and networks grew, the need for forensic capabilities to investigate cybercrime and preserve digital evidence became clearer.
- By 3 November 2011, CSM's digital forensic lab became the first in the Asia-Pacific region to be accredited by the ASCLD/LAB for "Digital & Multimedia Evidence" discipline (based on ISO/IEC 17025:2005)

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The History of Computer Forensics in Malaysia (Cont.)

Growth of Services & Recognition

- CSM's Digital Forensics Department (DFD) expanded to cover multiple domains: computer forensics, mobile forensics, multimedia, embedded devices, cloud/big data
- CSM's analysts were gazetted as expert witnesses under Criminal Procedure Code 399 starting 23 February 2009, enabling their evidence and testimony to be admissible in Malaysian courts.
- The laboratory officially announced digital forensics services around 2002
- By 2010, the DFD had handled hundreds of cases annually for law-enforcement and regulatory agencies.
- In December 2016, CSM launched two forensic-tool products (Kloner & Pendua) to support crime-scene duplication and digital evidence preservation.
- In September 2025, CSM launched Malaysia's first **vehicle forensics laboratory**, to strengthen investigations involving vehicles in cross-border crimes, smuggling, human trafficking.

Agencies Involved in Computer Forensics Investigations

- Department of Defense
- Federal Bureau of Investigation
- U.S. Internal Revenue Service
- United States Secret Service
- Federal Law Enforcement Training Center
- National White Collar Crime Center
- INTERPOL
- Computer Technology Investigators Network
- InfraGard

Agencies Involved in Computer Forensics Investigations in Malaysia

Agency / Organization	Main Role in Computer Forensics	Key Functions / Responsibilities
Royal Malaysia Police (RMP / PDRM)	Primary law enforcement body handling digital crimes	• Conducts forensic analysis on seized digital devices • Operates Cyber Crime Investigation Division (CCID) • Collaborates with MCMC, CyberSecurity Malaysia, and Interpol
CyberSecurity Malaysia (CSM)	National cybersecurity specialist agency under MOSTI	• Provides digital forensic services via Digital Forensics Department • Manages Malaysia Computer Emergency Response Team (MyCERT) • Offers training, R&D, and incident response support
Malaysian Communications and Multimedia Commission (MCMC)	Regulator of communication and multimedia industries	• Investigates cyber offences under the CMA 1998 • Supports evidence collection and digital trace analysis • Coordinates with ISPs and other regulators
Malaysian Anti-Corruption Commission (MACC / SPRM)	Investigates corruption cases, including digital evidence	• Conducts forensic imaging and data recovery from electronic devices • Analyzes digital trails in corruption and abuse-of-power cases
Royal Malaysian Customs Department	Handles digital evidence in smuggling and tax evasion cases	• Uses forensic tools to trace illegal online trade and digital financial transactions
Bank Negara Malaysia (BNM)	Central bank with cyber forensics capability	• Investigates financial fraud, money laundering, and digital payment crimes • Oversees financial sector cybersecurity standards
Attorney General's Chambers (AGC)	Provides legal direction for evidence admissibility	• Ensures digital forensic evidence meets legal requirements under the Evidence Act 1950
Malaysian Armed Forces (MAF) / Cyber Defence Operation Centre (CDOC)	Focuses on military and national cyber defence	• Conducts forensic analysis of cyber attacks targeting national security infrastructure

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Summary

- Reviewed the definition and importance of digital forensics
- Reviewed different types of digital evidence
- Reviewed the skills required to become a digital forensics investigator
- Reviewed the history of digital forensics
- Reviewed agencies in the United States, internationally, & Malaysia involved in digital forensics investigations

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